Theory-based priors for the output gap

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Abstract

The estimates of potential output and the output gap can have strong policy implications, yet estimation raises severe unresolved identification challenges. We develop a general econometric framework that improves estimation by using theory-consistent models to inform the prior beliefs of a new flexible time series method. The approach nests standard unobserved-components models, while allowing external quantitative or qualitative information to be incorporated in a transparent and precision-weighted manner. Simulation evidence suggests that our prior can recover potentially asymmetric cyclical evolutions without having to estimate potentially challenging non-linear models. We find that the output gap was positive before the Great Financial Crisis, and less negative during Covid compared to the official CBO estimates. At present, we estimate the output gap to be small and negative.

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1 Introduction

Estimating potential output and the output gap correctly is central to macroeconomic policy. The distinction between weak demand and weak productive capacity shapes the interpretation of inflation, labor-market slack, and the appropriate stance of monetary and fiscal policy. However, this distinction is hardest precisely when it matters most: after large shocks, when potential output may itself shift and real-time estimates are uncertain.

Several debates in both the United States and the euro area illustrate that output-gap measurement is not a purely technical exercise, but a policy-relevant topic with deep theoretical foundations and strong policy implications. Before the Great Financial Crisis, the debate centered around whether the strong output growth was sustainable because driven by a strong potential and could be supported by expansionary monetary policy, or was more cyclical and potentially inflationary, hence suggesting a more cautious policy stance (Barigozzi and Luciani, 2023, Borio et al., 2013, Hasenzagl et al., 2022). During the Covid-19 pandemic, the debate was whether the slowdown in output was consistent with excessive slack and hence compatible with a strong monetary and fiscal policy, or whether it was structural, making expansionary demand policies potentially highly inflationary (Edelberg and Sheiner, 2021, Summers, 2021, Blanchard, 2021, Bianchi et al., 2021).

Because neither potential output nor the output gap is directly observable, their measurement hinges on identifying assumptions, which in turn requires prior beliefs. At a broad level, the literature offers two widely used approaches. The production-function approach defines potential output as the level of output consistent with current technologies and normal utilization of capital and labor, typically constructing it from estimates of trend inputs and total factor productivity (Fleischman and Roberts, 2011). The statistical approach—most prominently unobserved-components (UC) models—defines potential output as the stochastic trend of GDP (Jarocinski and Lenza, 2018). There are other structural approaches, such as DSGE models, but in practice most policy institutions rely primarily on either production-function or statistical decompositions. These approaches differ mainly in how identifying restrictions on trend and cycle are imposed. See Canova (2025) for a comprehensive analysis of alternative existing methods. As confirmed by the existing literature, prior beliefs are crucial in shaping the estimation of the latent variables of interest.

This paper develops a general time series econometric framework that improves the estimation of potential output and the output gap by bringing into the analysis a broader set of prior beliefs relative to what currently offered by existing methods. The key innovation is to introduce identifying beliefs directly at the level of the joint distribution of the latent trajectory, rather than through a particular state evolution equation. Since a prior is formed directly on the full vector of the trend and cycle, several prior beliefs can be naturally introduced by setting the prior moments accordingly. For instance, as we do in our application, we can inform the prior correlation structure of the model using the Smets and Wouters (2007) model. Other specifications are possible, for instance shaping the prior beliefs via the subjective beliefs entertained by policy makers. In so doing, our method offers a bridge between the existing literature and offers a flexible and transparent tool to widen the set of beliefs used to uncover the true, unobserved latent variables.

Our formulation is very general and nests standard unobserved-components models as a special case: conventional specifications such as random-walk trends or local linear trends correspond to particular covariance structures implied by state-space transition equations. Our approach therefore generalizes the UC framework by decoupling identification from a specific parametric law of motion and instead allowing restrictions to be imposed directly on admissible trajectories. However, our specification is more transparent, because it can analytically decompose any update in prior beliefs into each feature of the prior beliefs that contribute to the estimation. In addition, it allows for the use of the data to estimate the signal-to-noise ratio of the prior beliefs of the model, relative to the measurement error. Although we focus on potential output and the output gap, the proposed framework applies more generally to the estimation of latent "star" variables, providing a unified approach to incorporating external information into trend–cycle decompositions.

To assess the properties of the proposed framework, we first conduct a set of simulation experiments. First, we show that, indeed in this class of model, prior beliefs matter considerably when estimating potential output and gaps from the data. This suggests that there is value in retaining as much flexibility in the prior beliefs as possible, as made possible within our framework. Then we show that our Gaussian prior specified directly on the latent variable of the output gap can uncover asymmetric business cycle fluctuations without having to estimate a non-linear model featuring asymmetries, which would be computationally more demanding. Rather, it only requires simulating from a model featuring non-linearities in order to compute the mean and variance of the prior on the cycle. By comparison, the standard AR(2) prior with oscillatory parameters performs poorly in recovering the true unobserved business cycle fluctuations.

We then apply the framework to U.S. real GDP. We use our methodology to study three episodes: the strong output growth before the Great Financial Crisis, the drop in output during the Covid period, and the most recent period. For each period, we inspect to what extent the output dynamics are due to variations in potential output, changes in the output gap, or a combination of the two. We use our method to inform the time series model using the Smets and Wouters (2007) model. We use the model to inform the prior correlation structure, and use the marginal likelihood of the data to estimate how much weight to give to potential output, the output gap, or the measurement error.

Our findings are as follows. From 1960Q1 until 1992Q2, our estimates of the output gap are remarkably similar to the official estimates from the Congressional Budget Office (CBO). From 1992 until the latest period, some significant differences emerge. Before the Great Financial Crisis, the CBO estimated a negative output gap (-2.2% in 2002Q4), which is consistent with expansionary monetary policy. We find that the output gap was largely positive during the full expansionary period before the crises (up to 3%). During Covid, the CBO interpreted the large drop in output as a cyclical factor, a finding consistent with expansionary fiscal policy. We find that the output gap was negative but much smaller than indicated by the CBO (approximately -3% instead of -9%). At present, we estimate the current out output gap at -1%, although uncertainty is such that, contrary to the pre-GFC and Covid periods, the output gap could be zero.

The paper relates to a vast literature. Several papers have documented the standard properties of business cycle dynamics, dating back to Burns and Mitchell (1946), and including more recent work, for instance Stock and Watson (2002) and Stock and Watson (2013). Our method provides a way to introduce the qualitative business cycle features from this literature into a tractable joint distribution. As, for instance, Grant and Chan (2017), Planas et al. (2008), and Hasenzagl et al. (2022) we take a Bayesian approach to output gap estimation, although a large frequentist literature explores very related concepts (Harvey, 1985). We use precision sampling for inference (Chan and Jeliazkov, 2009). More broadly, our method can be applied to any 'star' decomposition, including for instance to r star (Del Negro et al., 2017, 2019). Last, several work has been done to shape prior distributions for time series methods using theory-based information, see for instance Ingram and Whiteman (1994), Del Negro and Schorfheide (2004) and Renzetti (2025). The contribution of our paper is similar in spirit, and works in the field of latent variable estimation. Technically, the paper also relates to Loria et al. (2022) in using simulations to inform prior moments.

The remainder of the paper is organized as follows. Section 2 presents the econometric framework and establishes its relationship to standard state-space representations. Section 3 reports simulation evidence. Section 4 applies the methodology to U.S. GDP and compares the resulting estimates with conventional approaches. Section 5 concludes.

2 The model

We consider the following unobserved component model for log GDP, y_t:

$$y_t = \tau_t + g_t + u_t, \qquad u_t \sim \mathcal{N}(0, \sigma_u^2), \tag{1}$$

where τ_t is the trend component, g_t is the cyclical component—the output gap—and u_t is a measurement error term, assumed to be Normally distributed with mean zero and variance σ_u^2. The analysis is presented in a univariate setting for expositional clarity, but the framework can be extended to multivariate unobserved-component specifications.

Our objective is to estimate τ_t and g_t, namely output trend and the output gap, from the observed data on y_t. This involves a basic identification challenge: without additional assumptions, the two latent components cannot be separately identified, since the data only inform their sum. To see this, write the model in compact form as

$$\boldsymbol{y} = \boldsymbol{m} + \boldsymbol{u}, \qquad \boldsymbol{u} \sim \mathcal{N}(0, \sigma_u^2 \boldsymbol{I}), \tag{2}$$

where $\boldsymbol{m} = \boldsymbol{\tau} + \boldsymbol{g}$. For any given path of $\boldsymbol{m}$, infinitely many decompositions into $\boldsymbol{\tau}$ and $\boldsymbol{g}$ are equally consistent with the observed data. A higher trend combined with a lower cycle, or a lower trend combined with a higher cycle, implies the same $\boldsymbol{m}$ and therefore the same path for output. Separating $\boldsymbol{m}$ into potential output and the output gap thus requires additional structure.

Within a Bayesian framework, this structure is provided by prior laws of motion imposed on the two latent components. (Stock and Watson, 2002; Planas et al., 2008; Stock and Watson, 2013; Grant and Chan, 2017; Jarocinski and Lenza, 2018; Hasenzagl et al., 2022). These priors encode beliefs about their dynamic properties and are central to posterior inference. Typically, the trend is modeled as a slowly evolving process, reflecting the idea that potential output changes gradually over time, while the cycle is modeled as stationary and mean-reverting, capturing short- to medium-run fluctuations around trend. Identification is therefore achieved by imposing distinct dynamic structures—expressed as different priors—on τ_t and g_t.

2.1 Our prior for the output gap

In this section, we introduce beliefs about the output gap that go beyond those implied by purely statistical prior laws of motion for the output gap and output trend. The key idea is to enrich the prior structure so that additional, potentially theory-driven information can enter the estimation. This is useful because business cycle models often imply properties of cyclical fluctuations that the researcher may want to preserve when estimating the output gap, such as their typical duration, degree of persistence, oscillatory behavior, or possible asymmetries between expansions and recessions. More generally, theory may also suggest state-dependent dynamics or interactions between trend and cycle that are difficult to represent through simple conditional laws of motion.

These considerations motivate a more flexible prior specification, in which economic beliefs enter directly through the joint distribution of the latent trend and cycle. By specifying a joint prior over the full paths of the trend and cyclical components, $(\boldsymbol{\tau}, \boldsymbol{g})$, the framework can encode rich dynamic restrictions, including the autocorrelation structure of each component and cross-correlations between trend and cycle at different horizons. Specifically, we assume that

$$\begin{pmatrix} \boldsymbol{\tau} \\ \boldsymbol{g} \end{pmatrix} | \sigma_u^2 \sim \mathcal{N}\left(\boldsymbol{\mu}, \sigma_u^2 \mathbf{\Lambda R \Lambda}\right), \tag{3}$$

$$\sigma_u^2 \sim IG(a, b), \tag{4}$$

where

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_\tau \\ \boldsymbol{\mu}_g \end{pmatrix}, \qquad \mathbf{R} = \begin{pmatrix} R_\tau & R_{\tau g} \\ R'_{\tau g} & R_g \end{pmatrix}, \tag{5}$$

and

$$\mathbf{\Lambda} = \begin{pmatrix} \lambda_\tau \boldsymbol{I}_T & 0 \\ 0 & \lambda_g \boldsymbol{I}_T \end{pmatrix}. \tag{6}$$

Equivalently,

$$\begin{pmatrix} \boldsymbol{\tau} \\ \boldsymbol{g} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_\tau \\ \boldsymbol{\mu}_g \end{pmatrix}, \sigma_u^2 \begin{pmatrix} \lambda_\tau^2 R_\tau & \lambda_\tau \lambda_g R_{\tau g} \\ \lambda_\tau \lambda_g R'_{\tau g} & \lambda_g^2 R_g \end{pmatrix} \right). \tag{7}$$

The prior mean vector $\boldsymbol{\mu}$ and the correlation matrix $\mathbf{R}$ summarize the information that enters the model from economic theory or from a structural model. In particular, $\boldsymbol{\mu}_\tau$ and $\boldsymbol{\mu}_g$ denote the prior mean paths for the trend and the cycle, while the blocks of $\mathbf{R}$ describe their dependence structure. The diagonal blocks R_τ and R_g capture the autocorrelation structure of the trend and the cycle, respectively, whereas the off-diagonal block $R_{\tau g}$ captures cross-correlations between the two components at different dates.

In practice, $\boldsymbol{\mu}$ and $\mathbf{R}$ may be available in closed form or generated through simulations from a structural model. This situation often arises when the structural model is too complex to estimate but easy to simulate. In either case, they remain directly linked to economically interpretable parameters. Different structural assumptions therefore imply different prior mean paths and correlation structures, and hence different priors over the latent states in the unobserved-components model.

The parameters λ_τ and λ_g determine the scale of the prior variation of the trend and the cycle, respectively. Since the prior covariance matrix is scaled by the measurement-error variance σ_u^2, these parameters govern the variability of each latent component relative to the noise in the measurement equation. Thus, λ_τ, λ_g, and σ_u^2 determine the scale of the prior covariance,

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\mathbf{R}\boldsymbol{\Lambda} = \begin{pmatrix} \boldsymbol{\Sigma}_\tau & \boldsymbol{\Sigma}_{\tau g} \\ \boldsymbol{\Sigma}'_{\tau g} & \boldsymbol{\Sigma}_g \end{pmatrix}. \tag{8}$$

Since $\boldsymbol{m} = \boldsymbol{\tau} + \boldsymbol{g}$, the joint prior over $(\boldsymbol{\tau}, \boldsymbol{g})$ also implies a prior distribution for the latent output component $\boldsymbol{m}$. In particular,

$$\boldsymbol{m}|\sigma_u^2 \sim \mathcal{N}\left(\boldsymbol{\mu}_m, \sigma_u^2 \Sigma_m\right), \tag{9}$$

$$\boldsymbol{\mu}_m = \boldsymbol{\mu}_\tau + \boldsymbol{\mu}_g, \tag{10}$$

$$\Sigma_m = \Sigma_\tau + \Sigma_g + \Sigma_{\tau g} + \Sigma'_{\tau g}. \tag{11}$$

This distribution summarizes the prior implications of the model for the component of output observed through the measurement equation, while the full joint prior determines how movements in $\boldsymbol{m}$ are decomposed into trend and cycle.

Our prior specification has several advantages. First, it is flexible: it allows information from a broad class of theoretical or structural models to enter the trend-cycle decomposition through the prior. Second, it is transparent and interpretable, since the prior objects $\boldsymbol{\mu}$ and $\mathbf{R}$ can be traced back to the structural assumptions or parameters that generate them. Third, it separates the identifying restrictions, encoded in the theory-implied mean and correlation structure, from the scale parameters λ_τ and λ_g, which determine the relative amount of variation attributed to trend and cycle.

Fourth, the joint-path formulation also allows the variation assigned to trend, cycle, and measurement error to change over time or around specific episodes. The Covid-19 period illustrates this: the extreme output movements of 2020 are hard to interpret with a time-invariant decomposition, since they may reflect a temporary cyclical shortfall, more persistent disruptions to potential output, or exceptional noise. In this framework, such views can be encoded in the prior by letting the relevant covariance-matrix entries vary around the Covid episode. For example, the prior may assign greater variance to g_t during the sharp contraction and rebound, greater variance to τ_t if the shock is viewed as persistent, or greater measurement-error variance if some movements are treated as noise. Thus, the framework accommodates episode-specific beliefs about how unusual output movements should be allocated across components while preserving the same unobserved-components structure.

2.1.1 Posterior inference

To see how the likelihood updates the prior, let's focus on the posterior on $\boldsymbol{m}$:

$$\boldsymbol{m}|\boldsymbol{y}, \sigma_u^2 \sim \mathcal{N}\left(\boldsymbol{\mu}_m^*, \sigma_u^2 \Sigma_m^*\right) \tag{12}$$

$$\boldsymbol{\mu}_m^* = \boldsymbol{\mu}_m + \Sigma_m(\Sigma_m + \mathbf{I})^{-1}(\boldsymbol{y} - \boldsymbol{\mu}_m) \tag{13}$$

$$\Sigma_m^* = \Sigma_m - \Sigma_m(\Sigma_m + \mathbf{I})^{-1}\Sigma_m \tag{14}$$

The posterior mean $\boldsymbol{\mu}_m^*$ is therefore a weighted average of the prior mean $\boldsymbol{\mu}_m$ and the observed data $\boldsymbol{y}$. The matrix $\Sigma_m(\Sigma_m + \mathbf{I})^{-1}$ plays the role of a signal-to-noise ratio: when prior uncertainty about $\boldsymbol{m}$ is large relative to measurement error, the posterior mean moves strongly toward the data; when prior uncertainty is small, the posterior remains close to the prior mean. Similarly, the posterior covariance Σ_m^* is smaller than the prior covariance Σ_m, reflecting the information about $\boldsymbol{m}$ contained in observed output.

The remaining question is how this posterior revision in $\boldsymbol{m}$ is allocated between the trend and cyclical components. This allocation is governed by the prior covariance structure between $\boldsymbol{\tau}$, $\boldsymbol{g}$, and their sum $\boldsymbol{m}$. In particular,

$$\begin{pmatrix} \boldsymbol{\tau} \\ \boldsymbol{g} \end{pmatrix} \mid \boldsymbol{y}, \sigma_u^2 \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_\tau^* \\ \boldsymbol{\mu}_g^* \end{pmatrix}, \sigma_u^2 \begin{pmatrix} \Sigma_\tau^* & \Sigma_{\tau g}^* \\ \Sigma_{\tau g}^{*\prime} & \Sigma_g^* \end{pmatrix} \right), \tag{15}$$

with posterior means

$$\boldsymbol{\mu}_\tau^* = \boldsymbol{\mu}_\tau + K_\tau(\boldsymbol{\mu}_m^* - \boldsymbol{\mu}_m), \tag{16}$$

$$\boldsymbol{\mu}_g^* = \boldsymbol{\mu}_g + K_g(\boldsymbol{\mu}_m^* - \boldsymbol{\mu}_m), \tag{17}$$

where

$$K_\tau = (\Sigma_\tau + \Sigma_{\tau g})\Sigma_m^{-1}, \tag{18}$$

$$K_g = (\Sigma_g + \Sigma_{\tau g}')\Sigma_m^{-1}. \tag{19}$$

The matrices K_τ and K_g determine how revisions in $\boldsymbol{m}$ are distributed across trend and cycle. They depend only on the prior covariance structure. If the prior implies that movements in $\boldsymbol{m}$ are mostly associated with trend variation, then K_τ is large relative to K_g, and the posterior revision is mainly assigned to $\boldsymbol{\tau}$. Conversely, if the prior attributes movements in $\boldsymbol{m}$ primarily to cyclical fluctuations, then the revision is mainly assigned to $\boldsymbol{g}$. Thus, while the likelihood updates the estimate of aggregate latent output $\boldsymbol{m}$, the prior determines how this update is decomposed into potential output and the output gap. The posterior covariances are

$$\Sigma_\tau^* = \Sigma_\tau - K_\tau(\Sigma_m - \Sigma_m^*)K_\tau', \tag{20}$$

$$\Sigma_g^* = \Sigma_g - K_g(\Sigma_m - \Sigma_m^*)K_g', \tag{21}$$

$$\Sigma_{\tau g}^* = \Sigma_{\tau g} - K_\tau(\Sigma_m - \Sigma_m^*)K_g'. \tag{22}$$

These expressions show that uncertainty about $\boldsymbol{\tau}$ and $\boldsymbol{g}$ is reduced only through the information that the data provide about their sum $\boldsymbol{m}$. The extent to which this reduction applies to each component again depends on the prior covariance structure.

2.1.2 Estimation algorithm

To estimate the model we propose the following approach. We calibrate λ_1, λ_2 by maximizing the marginal likelihood of the model, that is

$$p(\boldsymbol{y}|\boldsymbol{\Lambda}) = \int p(\boldsymbol{y}|\sigma_u^2, \boldsymbol{\tau}, \boldsymbol{g}) p(\sigma_u^2, \boldsymbol{\tau}, \boldsymbol{g}|\boldsymbol{\Lambda}) d\sigma_u^2 d\boldsymbol{\tau} d\boldsymbol{g} \tag{23}$$

Notice that the choice of λ_1, λ_2 helps explain the behavior of $\boldsymbol{y}$, not a better decomposition τ_t versus g_t. In other words, our approach selects the scale of latent variation relative to measurement error that best explains the observed series $\boldsymbol{y}$ giving the highest cumulative one-step-ahead predictive density for the observed data. Importantly this maximization in conditional on a given choice of $\boldsymbol{\mu}^*$ and $\boldsymbol{R^*}$, with the idea that given the theory based identifying assumption for disentangling τ_t from g_t, the researcher aims finding the best one step ahead predictive model.

Then, we draw from the posterior distribution of σ_u^2 and the latent states in $\boldsymbol{\tau}, \boldsymbol{g}$ by direct Monte Carlo, conditionally on the values of λ_1, λ_2 obtained in the maximization step. The direct Monte Carlo algorithm is made of two steps: in the first step the algorithm samples from the marginal posterior of σ_u^2. In the second step, it draws from the posterior distribution of $\boldsymbol{\tau}, \boldsymbol{g}$ conditionally on σ_u^2. The marginal posterior $p(\sigma_u^2|\boldsymbol{y})$ and the conditional posterior distribution $p(\boldsymbol{\tau}, \boldsymbol{g}|\sigma_u^2, \boldsymbol{y})$, obtained by simple Bayesian updating of the likelihood with the prior in (3) and (4), are as follows:

$$\sigma_u^2|\boldsymbol{y} \sim IG\left(a + \frac{T}{2},\ b + \frac{1}{2}\left[\boldsymbol{y}'\boldsymbol{y} + \boldsymbol{\mu}'\left(\boldsymbol{\Lambda R \Lambda}\right)^{-1}\boldsymbol{\mu} - \boldsymbol{\mu}_n'\boldsymbol{V}_n^{-1}\boldsymbol{\mu}_n\right]\right), \tag{24}$$

and the following conditional posterior for $(\boldsymbol{\tau}, \boldsymbol{g})$

$$\boldsymbol{\beta}|\sigma_u^2, \boldsymbol{y} \sim \mathcal{N}\left(\boldsymbol{\mu}_n, \sigma_u^2\boldsymbol{V}_n\right), \tag{25}$$

where

$$\boldsymbol{\mu}_n = \boldsymbol{V}_n\left[\boldsymbol{X}'\boldsymbol{y} + \left(\boldsymbol{\Lambda R \Lambda}\right)^{-1}\boldsymbol{\mu}\right], \tag{26}$$

$$\boldsymbol{V}_n = \left[\boldsymbol{X}'\boldsymbol{X} + \left(\boldsymbol{\Lambda R \Lambda}\right)^{-1}\right]^{-1}, \tag{27}$$

and $\boldsymbol{X} = \begin{bmatrix}\boldsymbol{I}_T & \boldsymbol{I}_T\end{bmatrix}$.

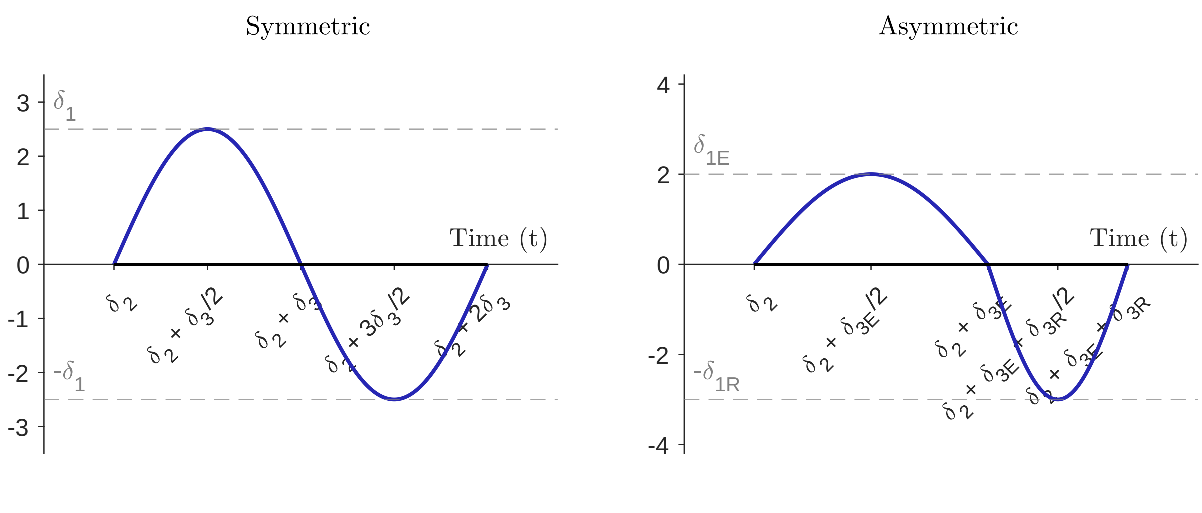
3 Simulation

A simulation on pseudo data helps illustrate that prior beliefs matter considerably in the estimation of latent variables, due to lack of identification in this class of models. This suggests that there is merit in retaining as much flexibility as possible in the formulation of prior beliefs, as argued in subsection 2.1.

We consider two datasets, both of size $T = 100$. For both datasets we simulate the true trend as the realization of a random walk with drift, using the same realization in order to ensure the same trend for both datasets. Starting from a random walk with drift model, we set $\tau_0 = 0$ and set μ such that the expected value of the trend equals 100 for the last period. We set $\sigma_\tau^2 = 0.05$ as the variance of the error term in the trend process.

The first dataset simulates the cycle as the realization of an AR(2) process. We set the parameters of the process to imply that the process is stationary, oscillatory, and with periodicity of 8 periods. This means that the response of the process to an impulse of positive sign takes 8 periods to complete a full expansion and then 8 periods to complete a full recession. The modulus of the oscillatory process is set to 0.90, which ensures that, leaving their periodicity unaltered, the cycles do not tighten too fast. The implied parameters are $\phi_1 = 1.6630$ and $\phi_2 = -0.81$. We then set $\sigma_g^2 = 0.2$ as the variance of the error term of the process.

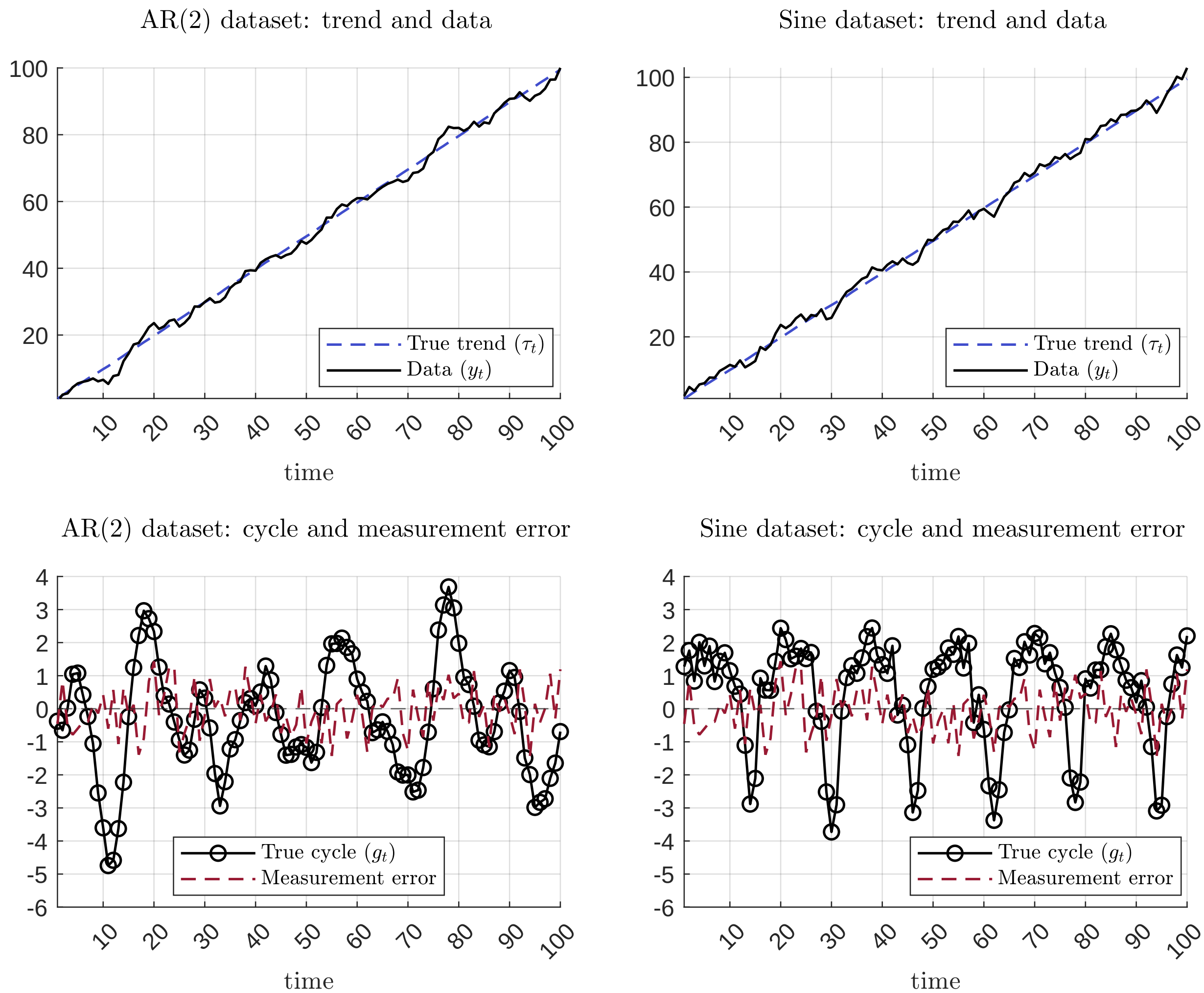
Figure 1: Sine functions



The second dataset aims to capture the widely held view that expansions and recessions are not symmetric phenomena. Expansions tend to be longer and shallower, while recessions can be abrupt, short and deep. To model this we use sine functions. We start from the symmetric function

$$\delta_1 \sin\left(\frac{t-\delta_2}{2\delta_3}\pi\right), \tag{28}$$

Figure 2: Simulated datasets



which equals zero at the generic period $t = \delta_2$, takes δ_3 periods to complete an expansion which peaks at $+\delta_1$, and then takes an additional δ_3 periods to complete a recession down to $-\delta_1$ (left panel of Figure 1). We generalize equation (28) to

$$\bar{g}_t = \begin{cases} \delta_{1E} \sin\left(\frac{t-\delta_2}{\delta_{3E}}\pi\right) & \delta_2 \leq t < \delta_2 + \delta_{3E}, \\ -\delta_{1R} \sin\left(\frac{t-(\delta_2+\delta_{3E})}{\delta_{3R}}\pi\right) & \delta_2 + \delta_{3E} \leq t < \delta_2 + \delta_{3E} + \delta_{3R}. \end{cases} \tag{29}$$

Starting at period $t = \delta_2$, an expansion takes δ_{3E} to complete and peaks at $+\delta_{1E}$, while a recession takes δ_{3R} to complete and reaches at $-\delta_{1R}$ (right panel of Figure 1). By construction, (29) converges to (28) for $\delta_{1E} = \delta_{1R} = \delta_1$ and $\delta_{3E} = \delta_{3R} = 2\delta_3$. In the second dataset we use equation (29) and set the true cycle equal to the realization for $T = 100$ periods from the process

$$g_t = \bar{g}_t + \delta_4 \; \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, 1), \tag{30}$$

with $\boldsymbol{\delta} = (\delta_{1E}, \delta_{1R}, \delta_2, \delta_{3E}, \delta_{3r}, \delta_4)'$ and

$$\delta_{1E} = 2\%, \quad \delta_{1E} = -3\%, \quad \delta_2 = 0, \quad \delta_{3E} = 12, \quad \delta_{3R} = 4, \quad \delta_4 = 0.2. \tag{31}$$

This parametrization implies that, compared to the AR(2) dataset, expansions last 4 periods longer and recessions last 4 periods shorter. The calibration of $(\delta_{1E}, \delta_{1R})$ ensures that also the magnitudes of the cycles are comparable across datasets.

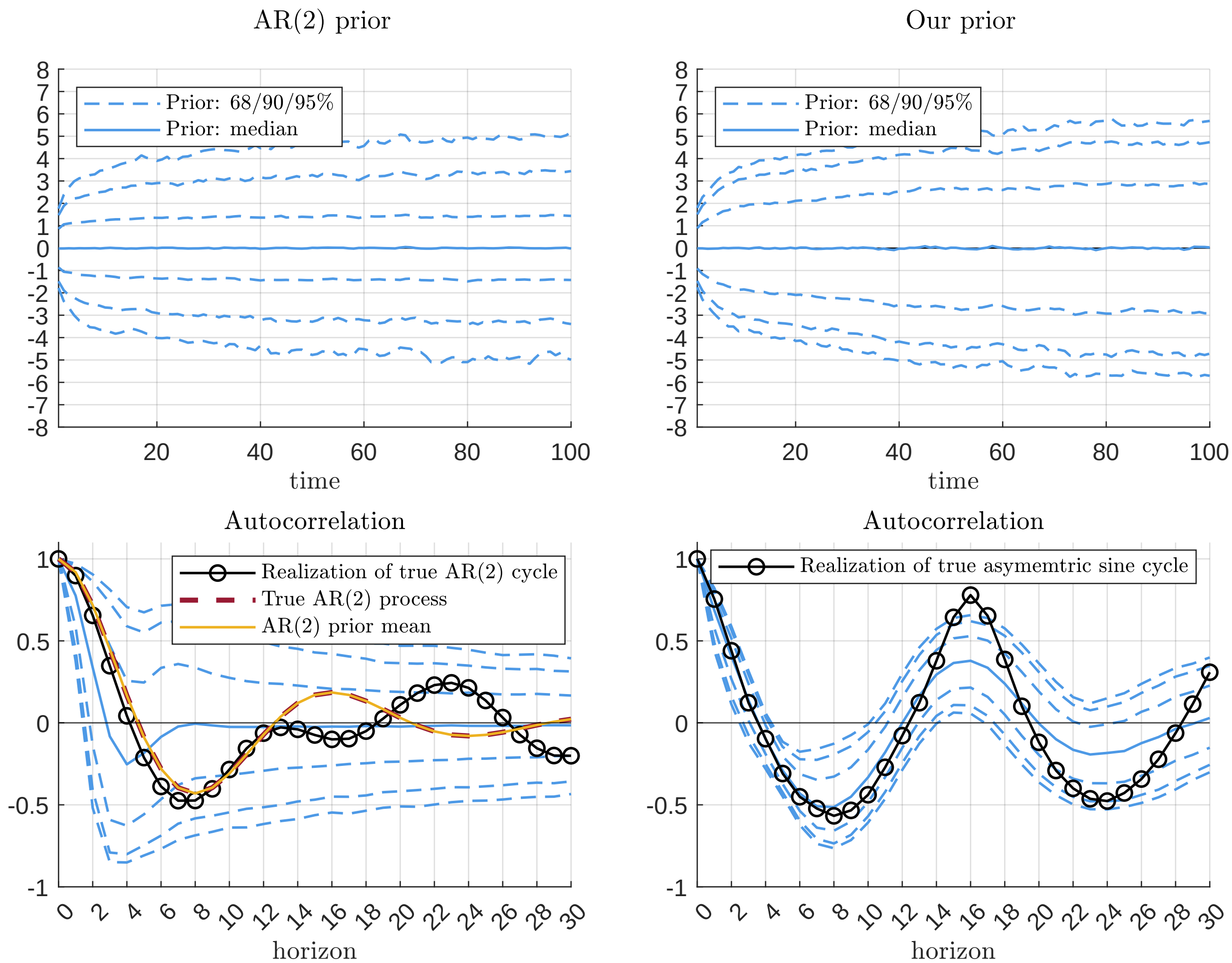
For each dataset, we compute the pseudo (log of) real GDP data as the sum of the true trend and the true cycle. We then add measurement error drawn from a Normal distribution with standard deviation 0.5, ensuring the same draws across datasets to improve comparability. Figure 2 shows the pseudo datasets. The left column refers to the dataset featuring an AR(2) cycle, while the right column refers to the dataset using the asymmetric sine cycle. The top panels show the true trend (blue dashed line) and the dataset (solid black line). The bottom panels show the true cycle (black circled lines) and the measurement errors (red dashed lines). For the exercise in this section it suffices to generate a single draw of each dataset. The Online Appendix reports a more comprehensive exercise with multiple draws of each dataset.

We estimate model (1) using, for both datasets, the same prior beliefs on the variance of the measurement error σ_u^2 and the same prior on the trend $\boldsymbol{\tau}$. For the former, we use an inverse gamma prior centered at 3 and with variance 2. For the latter, we use the random walk prior with drift. We use prior $p(\mu, \sigma_\tau^2, \tau_0) = p(\mu) \; p(\sigma_\tau^2) \; p(\tau_0)$, with a Normal prior for τ_0 centered at the first realization of the data and with variance 100, a Normal prior on μ centered at the value consistent with the ratio $\frac{y_T}{y_1}$ and with variance 0.5, and inverse gamma prior on σ_τ^2 centered at 0.03 with variance 0.005.

We then consider two alternative prior beliefs for the cycle. The first prior follows an AR(2) process, and is set to approximate the true AR(2) process of the first dataset. The joint Normal prior on (g_{-1}, g_0) is centered at zero with covariance 0.3 I_2. The joint Normal prior on (ϕ_1, ϕ_2) is centered at the true values, with covariance matrix $0.2 I_2$. The inverse gamma prior on the error term of the AR(2) prior is set to have mean 0.25 and variance 0.065.

The second prior we consider for the cycle is a prior informed by the asymmetric

Figure 3: Cycle: prior distributions



sine data generating process. One could in principle replace the AR(2) model-based prior with equation (29) and make inference on the hyperparameters $\boldsymbol{\delta}$. This approach is computationally more challenging, due to the nonlinearities involving equation (29) with respect to $\boldsymbol{\delta}$ and the lack of conditionally conjugate priors. Our proposed approach does not require sampling $p(\boldsymbol{\delta}|Y)$. Instead, it uses the asymmetric process (29) to *inform* a prior distribution that features a higher degree of tractability. We consider the prior distribution

$$\boldsymbol{g} \sim N(\boldsymbol{\mu}_g, \Sigma_g), \tag{32}$$

which is a simplified specification of our prior. We set $\boldsymbol{\mu}_g = \mathbf{0}$ to ensure that the first moment of the posterior is not biased towards the truth via the prior mean. As for

Σ_g, we set it as follows, using simulations from equation (29):

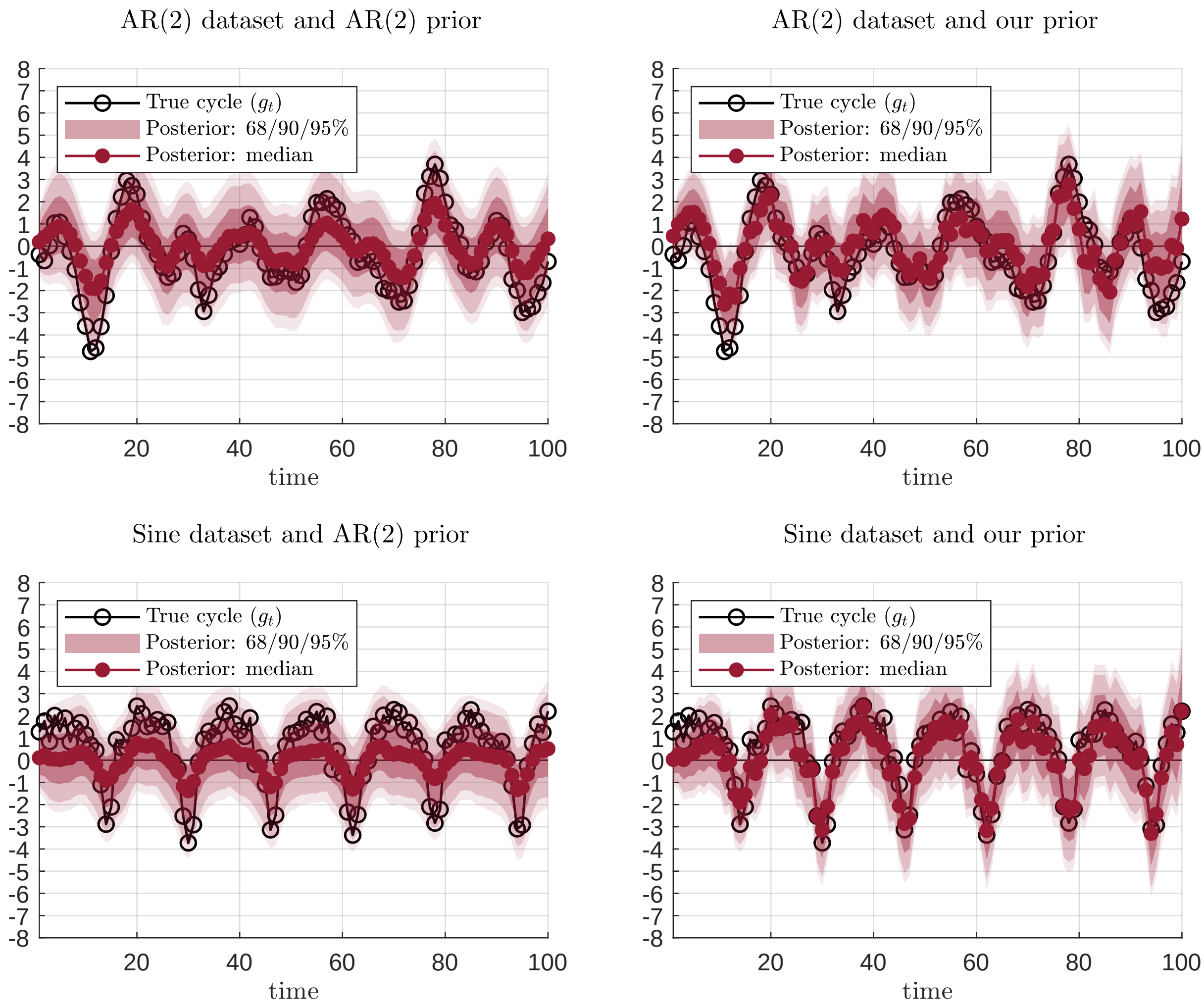
1. draw δ_{1E} from a continuous uniform distribution with support $[1.8, 2.2]$;
2. draw δ_{1R} from a continuous uniform distribution with support $[2.8, 3.2]$;
3. draw δ_2 from a discrete uniform distribution with support $[-20, 20]$;
4. draw δ_{3E} from a discrete uniform distribution with support $[10, 15]$;
5. draw δ_{3R} from a discrete uniform distribution with support $[1, 5]$;
6. set $\delta_4 = 0.02$
7. draw $\boldsymbol{g}$ from (29) given the above parameters $\boldsymbol{\delta}$;
8. repeat from step 1 a desired number of times;
9. store all draws. Set Σ_g equal to the empirical estimated covariance function as the average covariance across simulations.

The logic of the calibration of Σ is simple. We aim to incorporate into the prior the true belief that the data generating process generates asymmetric cycles. The randomlization on the parameters $\boldsymbol{\delta}$ is loose but centered around the true parameter values from the data generating process. This replicates the case in which, in applied work, the researcher holds an economic theory that reveals true information about the underlying unobserved output gap.

To further ensure comparability between the AR(2) prior and the asymmetric sine prior, we use the simulations described above to only construct the prior correlation structure rather than the prior covariance. We then compute the prior covariance to ensure that the marginal prior variances over periods $t = 1, .., 100$ are the same as with the AR(2) prior. This implies that the priors are approximately equally tight, but introduce potentially very different information via the correlation structure. This additional step does not affect the results considerably, but simplifies the illustration.

The top panels of Figure 3 report the pointwise prior median and bands associated with the two alternative prior distributions for the cycle. As it is clear from the figure, neither prior introduces a prior mean that reflects any cyclicality, hence neither prior biases the first moment towards the truth. The prior variance of the AR(2) is smaller at shorter horizons, consistent with the fact that the marginal prior for (g_{-1}, g_0) is not the marginal distribution implied by the AR(2) itself. By construction, the width of

Figure 4: Cycle: posterior distributions



the two priors is very similar, since only the correlation structure across time different across priors.

The lower panel of Figure 3 helps explore the difference in the two prior distributions by reporting a series of autocorrelation functions up to horizon 20. These are computed either on the realization of the true cycle, the realization of each prior draw, or the population autocorrelation function of the process. In the bottom left panel, the dashed red line shows the true autocorrelation function of the data generating process using an AR(2) cycle. This coincides with the yellow line, which shows the population autocorrelation of the AR(2) prior distribution for $\boldsymbol{g}$ evaluated at its mean. Since the prior mean for ϕ_1, ϕ_2 was set equal to the true values from the AR(2) dataset, the two lines coincide, and are very close to the autocorrelation function computed on the true realization of the cycle under the AR(2) dataset, marked in the black circled line. All three lines decrease for the first 8 periods and then increase for the subsequent 8 periods, reflecting the fact that it takes 8 periods for the AR(2) process to complete a full expansion or a full recession. Yet, the AR(2) prior fails to introduce this feature of the data despite a fairly tight prior on (ϕ_1, ϕ_2). This is clear from the blue lines, which report pointwise mean and percentiles that decease at horizon 4 rather than 8. The bottom right panel of Figure 3 shows the autocorrelations for our prior. The black circled line shows that the autocorrelation function of the true asymmetric sine cycle is very different from the AR(2) process. But contrary to the AR(2) prior, our prior introduces patterns that are consistent with the data generating process.

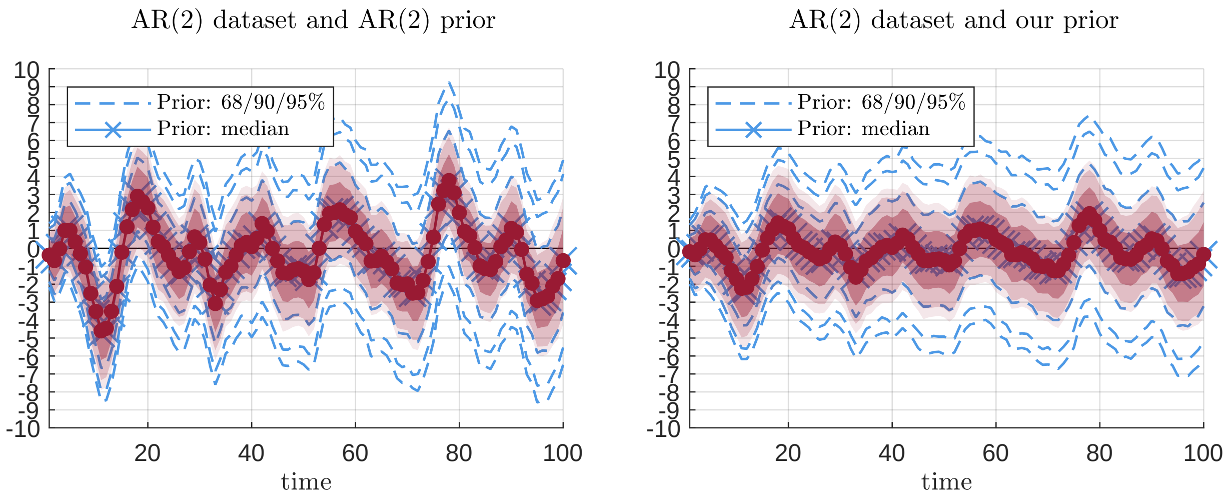
Figure 4 shows posterior distributions on the cycle (see the Online Appendix for the results for the trends). The figure reports pointwise median and bands associated with all four combinations of the two datasets and the two priors. The left column of Figure 4 reports the results using the AR(2) the prior. As is clear from the figure, inference is very successful when the underlying true cycle is indeed an AR(2) process. However, the results are less satisfactory when the true cycle is the asymmetric sine function, despite on average similar patterns between datasets. In fact, the bottom left figure shows that the sampler correctly infers only the qualitative feature of the cycles, uncovering the true pattern of expansions and recessions. Yet, it fails to capture the true cycle quantitatively, underestimating the cyclical dynamics by around 50%. The results are different for our prior, as shown on the right hand side of the figure. Our prior successfully recovers the true process under both datasets.

The results from Figure 4 suggest that prior distributions can have a significant role in the estimation of latent variables. Indeed, by construction, the true trend and the true measurement error was the same across datasets, hence, within dataset, the only factor driving the difference in the posterior is the difference in the prior distributions on $\boldsymbol{g}$. We further support this illustration with an additional exercise. Consider for simplicity only the AR(2) dataset. Replace the random walk prior with a joint Normal prior

$$\boldsymbol{\tau} \sim \mathcal{N}(\boldsymbol{\mu}_\tau, \Sigma_\tau), \tag{33}$$

where we set Σ_τ to feature the same prior correlation structure as the random walk prior used so far, with prior marginal variances for each period set equal to the average variance of the random walk prior (this parametrization is inconsequential for the illustration). For both cases below, set $\boldsymbol{\mu}_\tau = \boldsymbol{y} - \boldsymbol{\mu}_g$. For Case a), set $\boldsymbol{\mu}_g$ equal to the true cycle, while for Case b) set it equal to half of the cycle. The prior and posterior distributions are reported on the left and right panel of Figure 5. It is clear from the figure that in neither case the first prior moment is updated, as prior and posterior pointwise median coincide under both cases. The only effect of introducing the data to the analysis is to tighten uncertainty. The result is due to the lack of identification of trend and cycle. Since by construction the prior distributions on $\boldsymbol{\tau}$ and $\boldsymbol{g}$ sum up to the data, the data cannot update the prior means.

Figure 5: Cycle: illustration of lack of identification



The take away from both exercises is not that our prior is superior to the AR(2) prior. The take away is simple: prior distributions matter. This result is not due to the length of the dataset, but to the nature of the decomposition exercise. Because of this identification problem, it is important to carefully select prior distributions on $(\boldsymbol{\tau}, \boldsymbol{g})$, potentially combining statistical information with theory-based beliefs.

4 Application to the US data

We now bring the framework to the data. This section explains how a theory - based prior is constructed using the structural model of Smets and Wouters (2007), and then describes the data used in the estimation. We subsequently use the resulting specification to study the evolution of output trend and the output gap in the United States.

4.1 Prior from Smets and Wouters (2007)

In our empirical implementation, the theoretical model provides two key inputs: the prior mean vector $\boldsymbol{\mu}$ and the correlation matrix $\boldsymbol{R}$ for the joint path of the output trend and the output gap. These objects summarize the theory's implications for the level, persistence, and co-movement of the two latent components. The vector $\boldsymbol{\mu}$ determines their prior mean paths, while $\boldsymbol{R}$ captures the autocorrelation structure of each component and the cross-correlations between them at different leads and lags.

In log-linearized DSGE models, these moments follow directly from the state-space representation. Because the solution expresses endogenous variables as linear functions of the state vector, the unconditional means, variances, autocorrelations, and cross-correlations of output, the output trend, and the output gap can be computed in closed form. These moments depend on the model's deep structural parameters, including preferences, nominal and real rigidities, policy-rule coefficients, and the persistence and variance of structural shocks.

We use the Smets and Wouters (2007) model to discipline the prior through the theory-implied objects $\boldsymbol{\mu}$ and $\boldsymbol{R}$. The model provides a structural interpretation of the dynamics of output, its trend, and its cyclical component, making it a natural source of prior information for the output gap estimation. For any given calibration of the structural parameters, the DSGE model implies a corresponding prior mean vector and correlation matrix. Hence, $\boldsymbol{\mu}$ and $\boldsymbol{R}$ are treated as functions of the calibrated parameters. Alternative calibrations - reflecting different assumptions about nominal rigidities, habit formation, investment adjustment costs, monetary-policy responses, or shock persistenceâwould generate different values of $\boldsymbol{\mu}$ and $\boldsymbol{R}$, and therefore different priors for the latent components. The parameter values used in our baseline calibration are reported in Table 1.

4.2 Data

We estimate the evolution of potential real GDP and the output gap in the United States from 1960Q1 to 2025Q3. Our empirical analysis focuses on whether the declines in economic activity following the Great Financial Crisis and the Covid pandemic show evidence of a reduction in potential output, or whether these episodes are better interpreted as temporary shortfalls in actual output - that is, as widenings of the output gap.

4.3 Results

Figure 6 shows the heat map of the evaluation of the marginal likelihood of the model. The left panel shows the marginal likelihood evaluated in $(\lambda_\tau, \lambda_g)$, while the right hand side in (λ, δ). Our method selects $(\lambda_\tau = 15.65, \lambda_g = 1.21)$ and $(\lambda = 15.69, \delta = 0.99)$, which is consistent with the trend being on a wider scale than the cycle. Hence, the estimation of the model weights heavily on the trend, but attaches a non negligible prior variance to the cycle. This is an encouraging result, as the prior distribution only introduces dogmatically the correlations within and across trend and cycle, but not the variances. These are uniquely pinned down by the data via the marginal likelihood.

Figure 6: Marginal likelihood

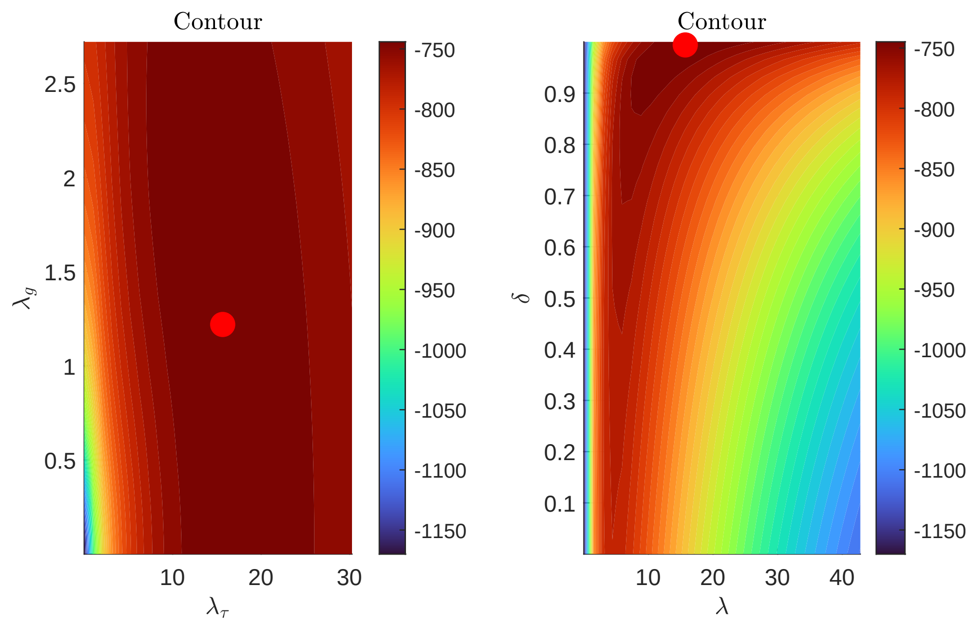
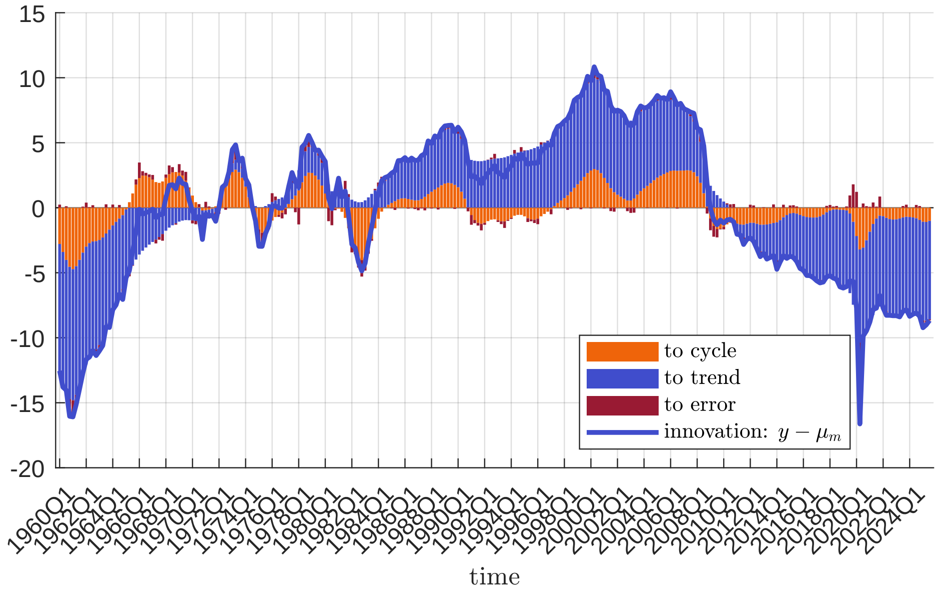


Figure 7 helps appreciate how the data update the prior distributions of the model. The blue line shows the innovation in the model, $\boldsymbol{y} - \boldsymbol{\mu}_m$. The red bars show to what extent the innovation is allocated by the model to the error term. The rest must be absorbed by a combination of trend and cycle. As indicated by the figure, a large component is allocated to an update in the mean of the trend. However, several periods call for a revision in the prior beliefs of the output gap, including soon before the Great Financial Crisis.

The results on the output gap are clearly displayed in Figure 8. The top panel shows the prior distribution. The width of the prior is pinned down by the estimation

Figure 7: Decomposition of update from the data



of λ_g, as explained above. The prior mean is set to zero. While the figure can illustrate the pointwise marginal prior distributions of the cycle over all periods, it hides a rich correlation structure. This is directly informed by the theoretical model used to inform the prior.

The lower panel of Figure 8 reports the results of the analysis. The solid red line shows the pointwise posterior median while the bands show the 68% to 95% pointwise bands. The solid green line shows the official estimate of the output gap according to the CBO. Last, the dashed red line shows the pointwise median value fo the residual.

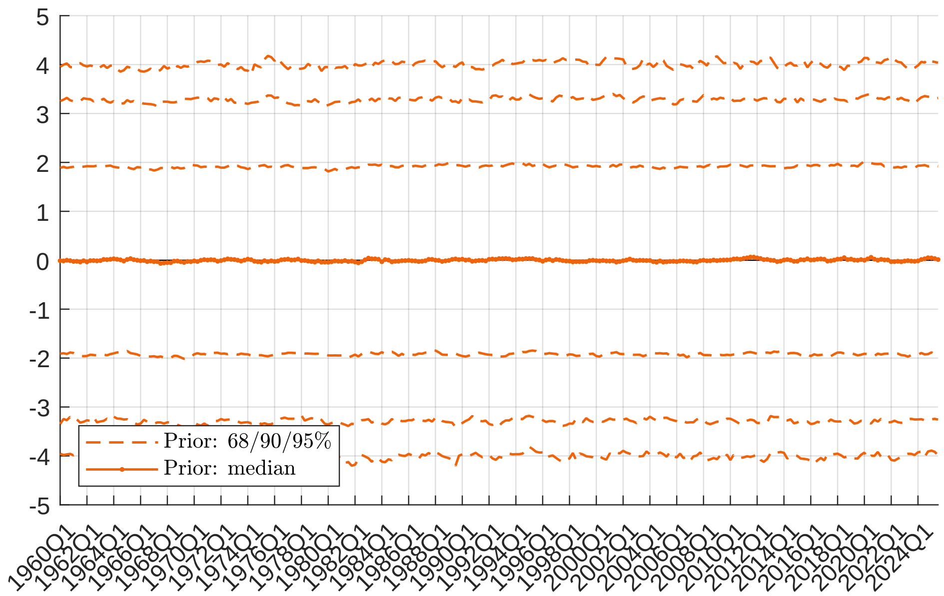
Several interesting results emerge from Figure 8:

- The model explains most of the data via the trend and the cycle, as only a limited amount of the innovation is absorbed by the measurement error. The only notable exception is during Covid. Our model is flexible enough to allow for part of the data to be explained by the measurement error. By contrast, the CBO allocates the full data to either trend or cycle;

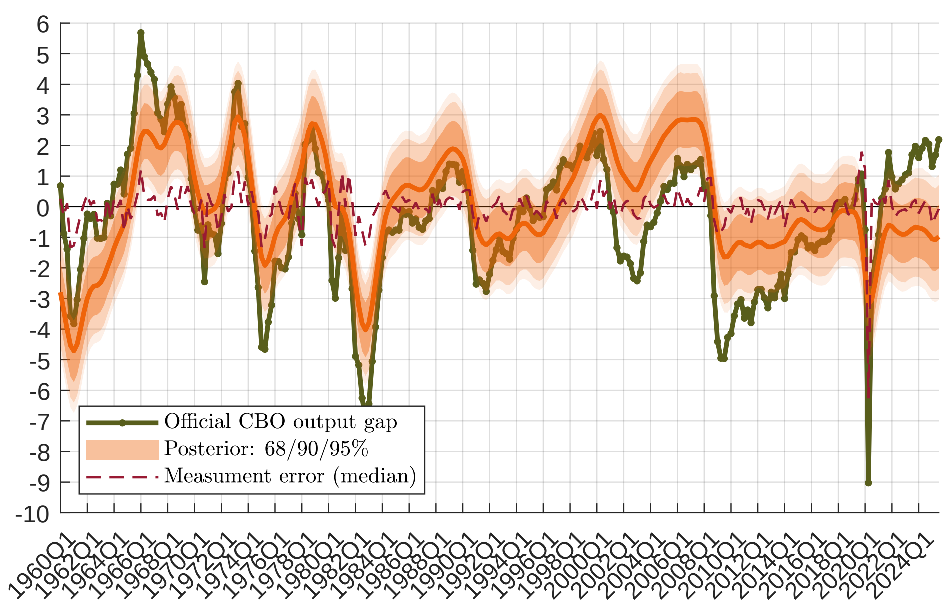
- Until 1992Q2, our estimates of the output gap are very in line with the official CBO estimate. This result is remarkable in that nothing in the model uses the CBO estimates to inform our prior beliefs.

Figure 8: Output gap

Our prior on the cycle



Posterior, measurement error, CBO estimate



- The CBO interprets the period before the Great Financial Crisis as a period with a sudden drop in the output gap, around 2004. This result was consistent with strongly expansionary monetary policy. Our estimates contradict this finding, in that the output gap is always positive. 2024 was a period of tighter output gap, which remained positive;

- The Covid crisis is interpreted by the CBO as fully driven by the cycle. Our method finds that part of the drop in output was due to a measurement error. our estimates of the output gap is -3% in 2020Q2, contrary to -9% from the CBO;

- At present, we estimate the output gap to be negative, although small.

5 Conclusions

Decomposing GDP into potential output and the output gap is fundamentally an exercise in identification. Because both components are latent, any empirical decomposition necessarily embeds restrictions on the joint dynamics of trend and cycle. The choice and strength of these restrictions have direct implications for inference and policy analysis.

This paper develops a general econometric framework that makes these restrictions explicit and continuously tunable. By specifying priors directly over the joint distribution of the latent trajectory, rather than through a particular state evolution equation, the framework nests standard unobserved-components models while enlarging the set of admissible dynamics. Identification is therefore decoupled from a specific parametric law of motion and instead implemented through mean and covariance restrictions on the full path of the latent variables.

The simulation evidence shows that our approach is sufficiently flexible to potentially introduce non-linear and asymmetric features into the prior moments, without having to estimate non-linear models. Rather, only simulation from such model is required to inform our prior moments. The empirical application shows the Great Financial Crisis was preceded by a positive rather than negative output gap. The Covid pandemic is interpreted by the model as parly due to a measurement error, partly a decline in output gap. .

Although the analysis of our paper focuses on potential output and the output gap, the framework applies more broadly to the estimation of latent "star" variables. Our framework can be extended to time-vaying paramer models in which economic theory informs the potential evolution of the parameters.

6 Appendix

6.1 Smets and Wouters prior: calibration

Table 1: Calibrated parameters: Smets and Wouters model

Parameter	Description	Value
δ	Depreciation rate	0.0250
ϕ_w	Gross wage markup	1.5000
$\bar{g}/\bar{y}$	Steady-state government spending share	0.1800
ε_p	Curvature of Kimball aggregator, prices	10.0000
ε_w	Curvature of Kimball aggregator, wages	10.0000
α	Capital share	0.2400
σ_c	Risk aversion	1.5000
ϕ_p	Fixed cost share / gross price markup	1.5000
ρ_{ga}	Feedback from technology to exogenous spending	0.5100
φ	Investment adjustment cost	6.0144
λ	External habit degree	0.6361
ξ_w	Calvo parameter, wages	0.8087
ι_w	Wage indexation	0.3243
σ_l	Labor supply elasticity parameter	1.9423
ξ_p	Calvo parameter, prices	0.6000
ι_p	Price indexation	0.4700
ψ	Capacity utilization cost parameter	0.2696
r_π	Taylor-rule inflation response	1.4880
ρ	Interest-rate smoothing	0.8762
r_y	Taylor-rule output-gap response	0.0593
$r_{\Delta y}$	Taylor-rule output-growth response	0.2347
ρ_a	Persistence of productivity shock	0.9977
ρ_b	Persistence of risk-premium shock	0.5799
ρ_g	Persistence of spending shock	0.9957
ρ_i	Persistence of investment-specific technology shock	0.7165
ρ_r	Persistence of monetary-policy shock	0.0000
ρ_p	Persistence of price-markup shock	0.0000
ρ_w	Persistence of wage-markup shock	0.0000
μ_p	MA coefficient, price-markup shock	0.0000
μ_w	MA coefficient, wage-markup shock	0.0000
$\bar{l}$	Steady-state hours	0.0000
$\bar{\pi}$	Steady-state inflation rate	0.7000
$100(\beta^{-1} - 1)$	Time-preference rate, percent	0.7420
$\bar{\gamma}$	Net trend growth rate, percent	0.3982

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